Physicist / Engineer physicist BSc, year III / semester 1
"Optics problem solving" – homeworks

Homework 1

Dr. Gábor Erdei, 12-09-2025

Given point P on the *z* axis in air, and on its right another medium of refractive index *n*. The interface between these two media is cylindrically symmetric to the *z* axis, its cross sectional profile can be described by curve g. By the help of Fermat's principle let us determine the equation of the g curve, by which the surface can image point P to infinity in the +*z* direction! The equation should not neccessarily be simplified, but if one can arrange the terms containing x^2 and z^2 to one side and the constant to the other, then what curve do we get?

